CYBERSHIELD AI

AI-Powered Cyber Safety Assistant

Mini Project Documentation

Project Website: https://polite-safe-shield-hub.base44.app

1. Project Overview

CyberShield AI is an AI-powered cyber safety platform designed to help students, families, senior citizens, and other non-technical users identify suspicious online activities, understand potential cyber threats, and take safe defensive actions. The platform acts as a first-level cyber safety assistant by analyzing suspicious messages, URLs, screenshots, emails, and online situations.

2. Problem Statement

Cyber threats such as phishing, fake websites, scam messages, fraudulent job offers, payment scams, and social engineering attacks are increasingly common. Many everyday users cannot easily determine whether an online message, link, email, or request is trustworthy. CyberShield AI addresses this problem by providing simple AI-assisted risk assessments and practical defensive guidance.

3. Objectives

Help non-technical users understand common cyber threats.

Identify suspicious patterns in messages and online content.

Provide simple and understandable risk assessments.

Guide users toward safe defensive actions.

Improve cybersecurity awareness among students and families.

Provide a centralized platform for cyber safety learning.

Maintain analysis history and threat reports.

Provide a foundation for future security-intelligence integrations.

4. Key Features

AI Cyber Safety Assistant: Conversational assistant for questions about phishing, account security, suspicious links, OTP requests, scams, and other everyday cyber-safety situations.

Scam Message Analyzer: Analyzes SMS, WhatsApp messages, social-media messages, email text, and other suspicious content for urgency, financial requests, credential requests, impersonation, suspicious links, and other warning signs.

URL Risk Analyzer: Assesses available website and URL signals such as HTTPS, URL structure, suspicious keywords, domain characteristics, and reputation information when available.

Screenshot Analyzer: Allows users to upload screenshots of messages, emails, websites, payment requests, or social-media content for AI-assisted analysis.

Phishing Email Analyzer: Examines sender characteristics, subject, content, links, credential requests, financial requests, urgency, and impersonation indicators.

Incident Response Assistant: Provides defensive guidance when a user has clicked a suspicious link, entered a password, shared an OTP, sent money, downloaded a suspicious file, or suspects account compromise.

Cyber Awareness Learning: Provides educational material covering phishing, social engineering, password security, MFA, safe browsing, scams, identity protection, and device security.

Cyber Awareness Quiz: Interactive cybersecurity questions with answers, explanations, and score/progress tracking.

Job & Internship Scam Detection: Helps students identify warning signs such as registration fees, upfront payments, suspicious recruiters, unrealistic offers, and pressure to act quickly.

Online Shopping Scam Detection: Highlights warning signs such as unusual discounts, suspicious URLs, missing business information, pressure to pay, and suspicious payment methods.

5. User Roles

5.1 Normal User

Use the AI assistant

Analyze messages, URLs, screenshots, and emails

Get incident-response guidance

View and save threat reports

View analysis history

Learn cybersecurity

Take awareness quizzes

5.2 Administrator

Manage users

Manage cybersecurity knowledge

Manage learning content

Manage quiz questions

View aggregated threat statistics

Manage threat categories

6. System Architecture

The application follows an AI-assisted defensive cyber-safety workflow:

User submits a message, URL, email, screenshot, or cyber-safety question.

CyberShield AI identifies the type and context of the request.

The appropriate analysis component processes the submitted information.

Available signals and suspicious indicators are evaluated.

A risk assessment and confidence level are generated.

The system explains why the content may be suspicious.

Safe defensive actions are recommended.

A threat report can be stored for later reference.

7. Technology Stack

Frontend: React.js, Vite, Tailwind CSS, responsive component-based UI

Application Platform: Base44 AI platform

Artificial Intelligence: Large Language Model (LLM), AI-assisted text analysis, conversational AI, and OCR/Vision capabilities where supported

Database: Base44 application database for users, analyses, reports, conversations, knowledge articles, learning content, and quizzes

Authentication: User authentication with role-based access

8. Risk Assessment

CyberShield AI uses risk levels to communicate the seriousness of detected warning signs. The result is a risk assessment and not an absolute guarantee that an item is safe or malicious.

HIGH RISK – Strong or multiple warning signs require immediate caution.

MEDIUM RISK – Suspicious indicators are present and additional verification is recommended.

LOW RISK – No major suspicious indicators were identified, but users should still remain cautious.

9. Threat Report

Threat Type

Risk Level

Confidence

Detected Indicators

Explanation

Recommended Actions

Date and analysis details

10. Dashboard

The user dashboard provides an overview of cyber-safety activity and can display total analyses, high/medium/low-risk counts, recent threats, analysis history, learning progress, and threat reports.

11. Security and Privacy

Use authentication and role-based authorization.

Keep sensitive API credentials out of the frontend.

Validate user input and uploaded files.

Apply appropriate file-size and file-type restrictions.

Keep user data isolated between accounts.

Avoid unnecessary storage of sensitive raw content.

Use aggregated/anonymized statistics for administrative analytics where appropriate.

12. Cybersecurity Safety Guardrails

CyberShield AI is designed as a defensive cybersecurity application. It supports threat awareness, risk assessment, account protection, incident response, and cybersecurity education.

The system does not provide instructions for unauthorized access.

It does not provide credential-theft or phishing instructions.

It does not provide malware creation instructions.

It does not provide authentication-bypass or exploitation instructions.

It does not recommend hacking back or retaliating against attackers.

13. Target Users

Students – protection from fake internships, scholarships, phishing, social-media scams, and fake job offers.

Families – protection from shopping scams, payment scams, fake support, impersonation, and suspicious links.

Senior Citizens – protection from fake bank calls, OTP scams, prize scams, and impersonation.

Common Internet Users – assistance with phishing, scam messages, suspicious websites, fraudulent offers, and social engineering.

14. User Workflow

Login/Signup → Dashboard → Select Analysis → Submit Content → AI Analysis → Risk Assessment → Explanation → Safe Action Guidance → Save/View Threat Report → History

15. Unique Selling Proposition

CyberShield AI is not designed as a generic AI chatbot. It is positioned as an AI-powered cyber safety platform that helps non-technical users identify online threats, understand the risk, and take safe defensive actions.

Core concept: Detect → Explain → Protect

16. Future Enhancements

Real-time threat intelligence integration

Advanced URL reputation services

Browser extension

Mobile application

Advanced RAG-based cybersecurity knowledge system

Personalized security recommendations

College cyber-safety portal

Family safety features

Multi-language support

Advanced AI-agent architecture

Automated security notifications

17. Project Website

Live Demo: https://polite-safe-shield-hub.base44.app

18. Project Information

Project Title: CyberShield AI

Project Type: Mini Project

Domain: Artificial Intelligence and Cybersecurity

Platform: Base44 AI

Primary Focus: AI-powered Cyber Safety

Target Users: Students, Families, Senior Citizens, and Common Internet Users

19. Conclusion

CyberShield AI aims to make cybersecurity more accessible to everyday users. Instead of requiring users to understand complex technical security concepts, the platform provides simple explanations, risk assessments, and safe defensive recommendations. By combining artificial intelligence, cybersecurity awareness, risk analysis, and an accessible user interface, CyberShield AI provides a practical approach to improving cyber safety in everyday digital life.

20. Project Link

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